

The seed germination model of enzyme catalysis

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The activity of enzymes and other biological macromolecules is often sensitively dependent on physiochemical context. Seed germination provides an analogy that helps to elicit the control and information processing capabilities of enzymatic networks. Like a seed, the enzyme takes a particular action (complexes with a specific substrate and catalyzes a specific reaction) when a specific set of milieu influences is satisfied. The context sensitivity, specificity and speed are enormously enhanced by the parallelism inherent in the electronic wave function (i.e. by the superposition of electronic states). This parallelism is converted to speedup through electronic-conformational interactions. The quantum speedup effect allows biological 'switches' to have qualitatively greater pattern recognition capabilities than electronic switches. Consequently the information processing and control capabilities of biomolecular systems exceed the capabilities obtainable from classical models and exceed the intuitive expectations that have developed through the study of such models.

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1. Introduction

Biological information processing is largely based on the recognition capabilities of macromolecules such as proteins and nucleic acids. Molecular shape plays an important role in these recognition processes. The term 'lock-key processing' is an over-simplification. But it provides a useful metaphor that distinguishes the fundamental character of biomolecular interactions from the type of switching behavior characteristic of digital computers and other electronic information processing systems.

Most models of enzymatic reaction systems treat the enzyme as a fairly simple switch that converts specific substrates to specific products under a well defined set of conditions. Such models are reasonably well formulated in terms of digital switching circuits, or by network dynamics whose control and information processing capabilities can, to a good degree of approximation, be captured by electronic analogs

(Capstick et al., 1992). The recognition capabilities are essentially masked at the level of network function, due to the fact that all the complex dynamics of recognition are summarized in a small set of reaction parameters. In effect, the dynamics are viewed as supporting the switching activity and as contributing to the global information processing capabilities only through this switching activity.

I shall argue in this note that the above point of view is inadequate and that pattern recognition, at the molecular level has system level implications that transcend the capabilities of putative electronic analogs. I shall argue further that the pattern recognition capabilities of enzymes cannot be characterized in terms of a well defined set of rate constants, but must instead depend on the detailed physics of the recognition process, including the quantum nature of submolecular processes.

2. The seed germination model

The reason for the breakdown of the electronic analogy is the context dependence of

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molecular recognition. Proteins, nucleic acids and various membrane components are highly sensitive to a variety of conditions, including control molecules and physiochemical features of the milieu. The phenomenon of seed germination provides a useful analogy. In many cases seeds remain dormant until a variety of conditions are met: temperature, moisture, presence of fire, chemical influences and so forth. The seed must make a judgement as to when it should enter an active stage in the face of an unknown and generally statistically variable environment. Once the decision to germinate is made the seed will mature into a plant in a reasonably definite way, subject to modulation by the environment. This of course assumes that it made a good decision. If it made a poor decision it could face an early freeze, suffer a growing season of insufficient length, or trigger its growth activity in an environment that is unsuited to it in terms of light or soil.

Enzymes, according to the seed germination analogy, trigger a specific reaction. But they do so only when a variety of conditions are met, that is, only in the proper context. The kinetics of enzyme reactions may of course be dependent on a number of factors, including both control molecules and milieu features. The seed germination image puts the emphasis on this modulation, and on the fact that a wide variety of conditions may have to be met for the enzyme to act at all. This is the feature that is most significant from the standpoint of information processing and control.

3. Contrast to conventional switching models

The above context sensitivity is an extension of the recognition capability. The difference from conventional switching models is that multiple influences control the switching activity. Simple switches recognize a small, well defined set of inputs (e.g. perform an 'or' function or an 'and' function). The capabilities of such switches can be completely specified with a small, well defined set of parameters that summarize all pertinent features of the interactions

responsible for switching. The recognition behavior of enzymes in the seed germination picture is not specifiable in such a simple manner. Characterizing the capabilities in this case requires a physics-based description that refers to the interactions responsible for recognition. Consequently these interactions have system level implications for control and information processing.

Why is it infeasible to characterize the context sensitive recognition capabilities in terms of a definite set of rate constants? Obviously it would be impossible to characterize human pattern recognition capabilities in this way, since this would require an unambiguous specification of the set of patterns recognized. The recognition capabilities of enzymes are of course not directly comparable to those of the organism as a whole and the 'intelligence' of the enzyme is minute by comparison. But the problem of specifying the set of circumstances under which an enzyme must act in an unambiguous way is qualitatively similar.

It is possible to express this difference quantitatively (Conrad, 1990). A system comprising n particles can support up to n^2 pairwise interactions. Many of the interactions may be irrelevant. For example, the significance of the interactions falls off with the distance between particles. This reduces the number of interactions relevant at any given time, though the number is still quite large. The number of possible interactions that would have to be considered over a span of time in a heterogeneous system in which the relative positions of the particles change is very much larger.

A switch in a digital computer can at most participate in a constant number of interactions, since its user manual definition must not change as more components are added. Suppose that a system comprising n particles is divided up into k homogeneous components, each containing m particles. The number of relevant interactions in any given component can be equated to the minimal number of interactions that could be used to satisfy the user manual definition in a reliable way. Let us denote this number by C_i . The interactions between components derives

from interactions between their constituent particles. This number must be a constant, due to the requirement that the addition of extra components must not compromise the user manual definition. Let us denote this constant by C_e . The number of allowable interactions is thus $C_e C_i k = C_e C_i (n/m)$. This is far less than the n^2 possible interactions in the system and in any practical situation it would be a relatively small number. Correspondingly, the patterns that digital switches recognize (OR gates, AND gates) are very simple; and more important the set of actual physical influences the switches are exposed to is very limited.

Electronic circuits of a more general nature may contain components (capacitors, inductors, resistors) that are subject to a greater range of conditions. Still the input-output characteristics of these components are subject to simple descriptions, reflecting the fact that the vast majority of detailed interactions within and between them are irrelevant. In short, the dynamical description of the components is not very much dependent on context. Decreasing the size of the components or increasing their density alters this. But from the standpoint of circuit theory, or connectionist models generally, the resulting sensitivity to context is equated to faulty behavior.

If the seed germination hypothesis is correct context sensitivity is a dominant feature. Modeling networks of enzymes by context insensitive elements could not then provide an adequate picture of their dynamics, or of their control capabilities.

4. Conformational processing

The seed germination model is a generalization of the lock-key model that takes the influence of milieu signals on conformation into account. Control molecules or other features of the milieu may serve as signals. The conformation of the enzyme changes in response to the signals or influences impinging on it. Different patterns of influence 'ripple' through the enzyme, yielding different shape features. When the conformation changes to a form that

satisfies particular conditions the enzyme recognizes a substrate and acts on it (or acts on an already bound substrate). In this way the enzyme recognizes, or categorizes, patterns of influence. Recognition of the substrate is an aspect of this broader recognition process.

Polymacromolecular complexes can also mediate this type of processing (see Fig. 1). Signals impinging on the complex trigger conformational changes of the constituent molecules. The conformational features of the complex change. The changed features trigger a catalytic action. Alternatively, an adaptor molecule may recognize a particular shape feature of the complex and link this to an enzyme that controls the output action. The intervention of the adaptor (or, more commonly, of a control site on the effector enzyme that serves as an adaptor) allows for an arbitrary association between the pattern of influences impinging on the complex and the action taken (Conrad, 1990). Once adaptors are introduced, it is possible to use a repertoire of adaptors to recognize different groupings of influence patterns and respond in an appropriate way. This is possible as long as some of the many shape features of the complex are common to different groupings of influence patterns.

5. Free energy minimization and natural selection

The essence of conformational processing is that it converts the problem of recognizing the pattern of impinging influences to a free energy minimization process. Thus all the clever physics of complex formation and self-assembly is brought to bear on what would otherwise be a difficult pattern recognition problem.

The problem is in general difficult due to the fact that the number of patterns of influence impinging on the enzyme or polymacromolecular processor grows in a combinatorially explosive way. If n influences are present at a given point in time and each influence is either present or absent, the number of different patterns of influence is 2^n . The number of ways of grouping the inputs grows as $2^{\frac{n}{2}}$. The number will be

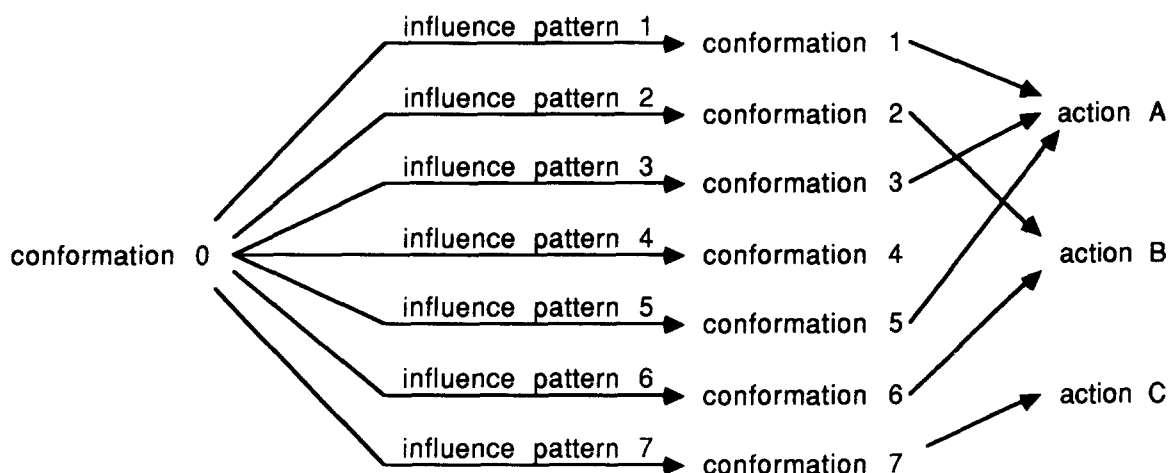


Fig. 1. Seed germination model of molecular information processing. The processor could be an enzyme or a polymacromolecular (supermolecular) complex. 'Conformation 0' is the initial conformation. The influence patterns (or contexts) are sets of milieu features, including physiochemical features and control molecules, along with the time intervals during which these features are present. Distinct influence patterns lead to different conformational forms of the complex, each characterized by some collection of shape features. The different conformations lead to different actions (including non-actions). If two conformations share a shape feature they can lead to the same action. Thus in the diagram conformations 1, 3 and 5 all lead to action A. This will happen if the common shape feature triggers the action of an enzyme that performs A. In this way the complex groups influence patterns into different categories. The shape features, hence the set of influence patterns, can be associated with actions in an arbitrary way. It is sufficient to adapt the control sites of effector enzymes to particular shape features. Alternatively, distinct adaptor molecules can serve to link the shape features of the complex to different effector enzymes. The number of different associations that the complex can mediate depends on the number of different or adaptor sites. The number of distinct influence patterns could of course be very much larger than indicated in the diagram.

greater if the influences each have a range of values and would increase enormously if the time of arrival has to be considered (it would increase to infinity if time does not admit a reasonable discretization).

This explosive growth in the search space provides another perspective on why it is difficult to characterize a powerful pattern recognition system in terms of a small number of parameters. Such a characterization would be possible if the significant inputs to the system are constrained to a small number (e.g. $n = 2$ or 3) and if the manner of grouping any remaining physical influences is such that there are only a very small number of functionally relevant outputs. But this is not in general the case. Whether a physical system takes an action or not, and how it acts, could conceivably be different in a functionally relevant way for a large fraction of the influence patterns.

If enzymes always (or nearly always) respond to two or three influences and no more, it would be possible to capture all their relevant behavior in a simple table. Such a restriction does not follow from physics, however. Very special 'engineering' would be required to create enzymes that satisfy such a simple and definitive functional specification. The process of evolution by variation and natural selection is the engineer. There is no reason to assume that this process works hard to restrict the pattern recognition capabilities of enzymes and polymacromolecular complexes for the convenience of human description. The more natural assumption is that natural selection exploits multiple weak interactions within enzymes, and between enzymes and their milieu, to enhance pattern recognition capabilities.

This does not mean that enzymes can keep up with the combinatorially explosive number of

groupings of influence patterns. This is not feasible and not necessary. What it means is that natural selection can mold enzymes to deal with much more challenging pattern recognition problems than electronic switching elements do, due to the fact that evolution is not restricted to creating products with convenient user manual definitions. Additionally the conformational properties of macromolecules, which allow for multiple interactions among many nuclei and electrons, afford much more intricate ways of integrating influences in space and time than does the much less complex structure of conventional electronic components.

6. Quantum speedup

Conformational processing, as intimated above, allows a pattern recognition problem to be mapped into a free energy surface. Natural selection does the mapping. The mapping can also be refined through adaptive processes within the organism. For example, cells that do not provide a suitable output may be subject to an error signal that alters their internal structure until a suitable intracellular arrangement of molecules is attained.

Searching a free energy surface for a minimum is somewhat like falling in a gravitational field. The enzyme solves the problem of recognizing the substrate and the context of influences by self-assembling ('falling') into the complexed state. The one difficulty is that falling to the desired minimum (perhaps the global minimum) might be impeded by being caught in local minima.

A classical system could never escape such local minima apart from thermal agitation, assuming that the energy surface is static. A quantum system could escape through barrier penetration. The atomic nuclei in the self-assembling complex are too massive to allow for such 'tunneling exploration' of the surface. The electron mass is small enough to allow for significant tunneling motions, however. Hydrogen bonds also have small enough mass to allow for some tunneling.

Interactions between the electronic and

nuclear coordinates would then allow for a quantum speedup of the search for a minimum. The quantum mechanical superposition principle is the fundamental feature. The superposition of electronic states provides an inherent parallelism that manifests itself as speedup. The interference of electronic states (components of the electronic wave function) of different energy leads to a changing probability distribution for the electronic coordinates, hence to a changing charge distribution. This is due to the fact that the changing phase relations among the different states allow for constructive and destructive interference (e.g. Bohm, 1951). The nuclei undergo a bobbing motion in response to the changing charge distribution. It is this bobbing motion, which in some instances self-amplifies due to a positive feedback relation with the electronic structure, that facilitates the jump to a new nuclear configuration, corresponding to a new minimum.

Could such a bobbing motion ever be significant as compared to thermal fluctuations? If the free energy surface were in fact static, thermal fluctuations would generally dominate at room temperature. The static picture, though a commonplace representation, is physically unrealistic. This is because of continual exchange of energy between electronic and nuclear coordinates and between kinetic and potential forms of energy (Conrad, 1979, 1991). A system of balls and springs could not be described by a static energy surface, due to the continual interchange between kinetic and potential energy (i.e. to the non-thermal motions of the system). Thus even in a classical balls and springs model of the enzyme-substrate complex the image of a static energy surface would be inadequate. The multidimensional energy surface would change with time, thereby opening and closing pathways between different minima. A classical system could escape local minima in this case, even in the absence of a sufficient enough thermal fluctuation, since the local minima would in many cases be temporary (apart from frictional damping of the non-thermal motions).

The quantum effect previewed above enor-

mously increases the importance of this continual opening up and closing of pathways and additionally is not subject to damping. Thermal fluctuations might be much larger than the quantum fluctuations, but they would not alter the structure of the energy surface. Consequently the relative importance of the two types of fluctuations cannot be assessed on the basis of the fluctuation energy alone.

7. Divergent and convergent dynamics

Suppose, as an extreme idealization, that we describe the self-assembling complex in terms of two quasi-independent wave functions that become correlated through mutual interactions. The wave functions could not in reality be independent, since the potential energies (and hence the Hamiltonians) of the electronic and nuclear subsystems are clearly not additive. Our assumption is that the interaction between the electrons and nuclei is sufficiently restricted that we can write down independent notational wave functions for the electronic and nuclear subsystems. If we can elicit a speedup effect under this condition, we could reasonably conclude that the effect would even be stronger if the interaction between electrons and nuclei is stronger.

For simplicity suppose that the complex has two accessible families of nuclear configurations, labeled by I and II, and two accessible families of electronic states, also labeled by I and II. The electronic wave functions can then be expressed as a superposition of the form

$$\Psi_e(x_e, t) = \sum_{r \in I} a_r(t) e_r(x_e) \exp(-iE_{e(r)}t/\hbar) + \sum_{s \in II} a_s(t) e_s(x_e) \exp(-iE_{e(s)}t/\hbar)$$

where x_e denotes the electronic coordinates, the e_r are eigenfunctions of the electronic system that are most naturally associated with nuclear configuration I, the e_s are eigenfunctions most naturally associated with nuclear configuration

II and the a_r and a_s are expansion coefficients. The $E_{e(r)}$ and the $E_{e(s)}$ are the electronic energies associated with the eigenfunctions e_r and e_s . Thus for simplicity we assume that the wave function can be split into time independent and time dependent parts. We also ignore ambiguous eigenfunctions that could be assigned to either minimum. This class would contribute to speedup and consequently ignoring it here does not weaken our argument.

Similarly the wave function for the collection of nuclei may be expressed as

$$\Psi_n(x_n, t) = \sum_{r \in I} b_r(t) n_r(x_n) \exp(-iE_{n(r)}t/\hbar) + \sum_{s \in II} b_s(t) n_s(x_n) \exp(-iE_{n(s)}t/\hbar)$$

where x_n denotes the nuclear coordinates, the n_r and the n_s are the eigenfunctions of the nuclear system when it is in conformation I and II, respectively and the $E_{n(r)}$ and $E_{n(s)}$ are the energies associated with these eigenfunctions.

The nuclear system, because of its large mass, should with high probability be assigned to a definite state. Thus if it is initially in conformation I all the b_s can for all practical purposes be set to zero. To make matters as unfavorable as possible to the speedup effect we can also assume that the nuclear configuration is initially in a single eigenstate, so that all the b_r apart from one can initially be set to zero. Practically speaking this is reasonable since the average values associated with the different n_r should peak very sharply due to the large mass. In reality the nuclear system is in a statistical ensemble of states and thus we could formulate our description in terms of density matrices. This would obscure matters, however, since the quantum speedup effect is due to the superposition of possible states that interfere with one another and not to the statistical distribution of wave functions.

The situation is different for the electronic system. For simplicity, and so as not to beg the question, we can assume that initially all the a_s

are negligible. Suppose, again to make matters as unfavorable as possible, that the electronic system is initially in a single eigenstate, so that only one of the a_r is initially non-negligible. If some of the electrons are not tightly bound to the nuclei, it will be possible for perturbations to induce transitions to eigenfunctions with energies close to that of the energy of the initial eigenfunction. Such perturbations are omnipresent, due to the interaction with the radiation field or due to either thermal or anharmonic motions of the nuclear system. The probability distribution of electronic coordinates will then be given by $p(x_e, t) = \Psi_e^*(x_e, t)\Psi_e(x_e, t)$, and therefore by

$$p(x_e, t) = \sum_r a_r^*(t)a_r(t)e_r^*(x_e)e_r(x_e) + \sum_r \sum_{r'} a_r^*(t)a_{r'}(t)e_r^*(x_e)e_{r'}(x_e) \exp\left(\frac{-i(E_{e(r)} - E_{e(r')})t}{\hbar}\right)$$

where $r \neq r'$. As noted earlier the probability distribution of the electrons is time dependent, due to constructive and destructive interference between eigenfunctions of different energy.

The electronic and nuclear wave functions must be consistent with one another. Self-consistent field theories of the Hartree-Fock type provide a useful formal framework for describing the evolution to a self-consistent state. The important difference is that in our case the evolution of a self-consistent field is used to represent in a conceptual way the self-organization of the complex rather than as a calculation scheme for approximating the wave function of a static molecular form. The justification is connected with the passage from a fully detailed microscopic description to a macroscopic description. The interacting molecules plus heat bath are described by some true time independent potential that governs both the electronic and nuclear dynamics. This potential is unknown and if it were known it would not be very useful since it would hide relevant

features in information about the heat bath. Some information must be discarded to bring out the essential features (always the case in passing from a microscopic to a macroscopic description). In each cycle through the scheme the potential functions and wave functions are updated. The physical significance of this is that new features of the true potential become relevant. A purely approximative use of a Hartree type scheme would not require the sequence of potentials to have any physical significance.

Details of this scheme are presented elsewhere (Conrad, 1992a, 1992b). Here it is sufficient to note that the electronic wave function determines the potential that governs the time development of the nuclear system; similarly the nuclear wave function determines the potential that governs the time development of the electronic system. The two wave functions (hence the two potentials) must be self-consistent and consequently the self-organization of the complex corresponds to the evolution to a self-consistent form (if a fully self-consistent form exists). As the electronic wave function changes (for example, due to perturbation by the changing nuclear coordinates or in response to the perturbing effects of the radiation field) the potential governing the nuclear motions will change (since the electronic charge distribution will change), leading to a change in the nuclear wave function. The potential governing the electronic system will then change, leading to a further change in the electronic wave function and therefore to further change in the potential governing the nuclear system.

Two basic types of relationships between the electronic and nuclear systems are possible:

(i). *Positive feedback.* In this case the nuclear and electronic wave functions are inconsistent. The a_r change with time, leading to a time dependence of the charge density. The nuclei undergo a slight bobbing motion in this changing charge density, thereby further spreading out the a_r . This increases the bobbing motion of the nuclei, again spreading out the a_r and so on. The effect is self-amplifying. The bobbing motion will either rapidly or slowly trigger a switch in the nuclear conformation, that is, a switch

from the b_r to the b_s . This switch will drag the electronic system along with it, corresponding to a major decrease in the a_r and a major increase in the a_s . The electronic system should then rapidly fall to the lowest energy state of the new nuclear conformation (apart from perturbations that again spread it out or that initiate a new divergent process).

(ii). *Negative feedback.* In this case the nuclear and electronic wave functions are inconsistent. Bobbing will occur, but it is insufficient to trigger a switch in the nuclear conformation. The slight changes in the conformation (really configurational changes) may occur, with associated slight change in the electronic wave function. The dynamics are convergent as long as the complex cycles within a given family of nuclear and electronic states. If the energy surface is viewed as static the system will appear to be jumping between subminima; if the surface is viewed as dynamically changing, then pathways between these subminima will appear and disappear.

The self-consistent state could either be a global or a local minimum. If it is a local minimum it will at most be metastable, due to the fact that values of the electronic coordinates that extend beyond the boundaries of self-consistency will always have a probability of occurrence, $p(x_e, t)$, that is non-zero. When these values occur the system may either switch to the new conformation directly, or a self-amplifying divergence may be initiated. When the complex does discover the global minimum it will remain in it, apart from thermal or quantum fluctuations that could not raise the energy permanently. However, it is also possible for the global minimum to be inherently unstable, in which case pathways for decomposition will open up dynamically after the complex is formed (Conrad, 1979). The complex will spontaneously break up in this case.

According to the above argument it should be possible for macromolecules to recognize each other with high specificity and at the same time rapidly assemble from a wide variety of initial contacts into a complex corresponding to a global minimum of the energy. Enzyme-

substrate complex formation, self-assembly of quaternary structures, and DNA annealing are examples. Molecular structures capable of assembling in this way may be a minority of all possible structures; but then biological macromolecules are also a very small, highly selected subset.

8. Physical correlates

Let us now consider more specifically the physical requirements that a molecular system would have to satisfy to yield the speed effect. These requirements are tantamount to empirical implications that could be used to test the applicability of the model.

i. *Connection between tunneling and superposition.* An electron described by a single eigenfunction would exhibit barrier penetration, due to the structure of the eigenfunction. However, tunneling through a barrier cannot be separated from superposition in a time dependent problem.

Suppose that the electron is initially associated with conformation I. Later it may be associated with conformation II as well as I (i.e. it is initially inserted into one well of a double well potential and later could be in either well). According to the uncertainty principle, $\Delta p \Delta x \approx \hbar$, Δp will decrease, due to the fact that Δx increases (the location of the electron is more uncertain). This means that the kinetic energy, $KE = p^2/2m$, must on the average decrease. (Alternatively, we can argue that in order to insert the electron into one of the wells initially we have to restrict Δx , leading to an initially higher kinetic energy. Or we can argue, from the time-energy principle, $\Delta E \Delta t \approx \hbar$, that increasing the time between insertion of the electron and measuring its energy allows the average energy to be smaller.) Thus even in the overly idealized case in which the electron is treated as being inserted into a given well of a double well potential and as initially in a single eigenstate of the double well, it will enter a lower energy state after enough time elapses for tunneling to occur. The transition could not take place if the electron were not in a superposition of states of different

energy, otherwise the spatial probability distribution of the electron could not change with time.

More generally, after initial insertion of the electron the expansion coefficients a_r are high and the coefficients a_s are low; later the a_r decrease and the a_s increase. Even if the basic functions chosen to describe the system are such that only one of the expansion coefficients is initially non-zero, this could not be the case at the end of the process (with the same choice of basic functions).

The above discussion points to the subtle connection between the form of the basic functions and their superposition as contributors to the speedup effect. The effect cannot be attributed to the form of the basic functions as long as one is working with a complete, orthonormal set of basic functions, since any such set will be adequate. However, if the basic functions are eigenfunctions the form would have physical significance, though no more significance than a properly constructed superposition of basic functions that individually have no physical significance. Superposition is physically necessary for speedup in both cases, since without superposition no time evolution would be possible. Our presentation of the physics behind the speedup effect has emphasized the superposition principle since the contribution from the form of the basic functions can be transformed from significant to insignificant, but the contribution from superposition is a *sine qua non* in either case.

(ii). *Role of hydrogen bonds.* As indicated already, the mass of the proton is small enough to undergo some tunneling. This is important since hydrogen bonds provide preferred pathways for electron tunneling in proteins (Beratan et al., 1990). We can picture the self-assembling system as comprising heavy nuclei, localized electrons, delocalized electrons and hydrogen bonds. Superposition of the delocalized electron states produces a bobbing motion of hydrogen bonds as well as of the other nuclei. However, the hydrogen bonds have a much greater likelihood of undergoing a transition to a new location than the heavy nuclei. This then

opens up new pathways for electron tunneling, thereby spreading out the superposition of electronic states (i.e. increasing previously small expansion coefficients). The interference effects increase, increasing the agitation of the heavier nuclei. The hydrogen bonds thus play an intermediating role in the self-amplification effect, both by virtue of their effect on electron tunneling and because the superposition of proton states can itself exhibit some spreading out. The important point is that evolution to self-consistency involves interactions among three mass scales (electrons, protons, heavy nuclei). Additionally, tunneling through hydrogen bond pathways increases the likelihood of degeneracies in the electronic configuration and therefore the likelihood of significant changes in the electronic wave function in response to small perturbations.

The role of hydrogen bonds provides a handle for experimental study of the speedup effect, due to the fact that hydrogen could be replaced by deuterium or tritium. This should substantially alter the intermediating role of hydrogen bonds and should therefore reduce the speed of complex formation.

(iii). *Inapplicability of the Born-Oppenheimer approximation.* As noted earlier, the electrons that contribute to the speedup effect should not be tightly bound to the atomic nuclei. Thus the (Born-Oppenheimer) assumption that the nuclei can be treated as fixed relative to the electrons cannot be the basis of a good description. In simple molecules, or in the case of tightly bound electrons, the small mass of the electrons allows them to relax very rapidly to a ground state in response to a nuclear motion; the influence of the electrons on the nuclei is very small and is ignored in the same fashion as the effect of a moving football on the motion of the earth might be ignored. The relative difference in mass is not the dominating factor in the case of the interactions between electrons and nuclei, however. Charge plays a role and is equal in the two systems. If the electrons are tightly bound the potential due to the charge can be combined with the potential due to the nuclei (because of the speed of relaxation). If the electrons are

delocalized, the relaxation will not be so fast and consequently the relative motion of electrons and nuclei will be important.

Models dealing with enzyme catalysis commonly consider the coupling of the electronic and nuclear degrees of freedom. The electronic-conformational interaction (ECI) model of Volkenstein is a prime example (1982). The nuclear and electronic potentials that determine the electronic energy levels change dynamically as the molecules dock, as nuclear degrees of freedom are frozen out in the course of docking, and as the charge clouds interact. The nuclear coordinates obviously change in the course of finding a complementary fit.

It might be supposed that the electronic coordinates readjust so fast that there is no chance of their influencing the nuclear coordinates and that Born-Oppenheimer could therefore provide the basis of an adequate description. Now imagine that we run the entire system (including heat bath) backwards in time. Then we would see the change in the electronic coordinates precede the change in the nuclear coordinates. If the dynamics allows this ordering to occur in the backwards direction of time it should also allow it to occur in the forward direction of time (because of the principle of microscopic reversibility). According to the model proposed in this paper this allowable forward effect bears on the rate of complex formation due to the fact that it subjects the nuclear coordinates to charge variations that arise from interference effects among stationary electronic states of slightly different energy.

iv. Absence of color and connection to infrared spectra. Biological macromolecules are colorless, apart from chromophores. Clearly the electronic states contributing to the superposition could not be so separated in energy that optical perturbations would be required to produce them. Infrared excitations are available, but these are normally attributed to vibrations and rotations of the atomic nuclei. Thus it might seem that superpositions of electronic states of different energy are precluded by the large energy gap between the ground state and the first excited states. This is not so for the non-Born-Oppenheimer electrons, however. These

electrons move relative to the nuclei and therefore some of the accelerations responsible for infrared radiation should be independently attributed to them. Since the whole system is a bound system this electronic contribution to the infrared spectrum, must be concomitant to transitions between closely spaced stationary states. In a fixed molecular structure this feature would be insignificant or absent, due to the fact that the motions of the nuclei are essentially periodic. Consequently the infrared spectrum is time independent. The spectrum changes during the formation of an enzyme substrate-complex, however, and some of this change should be attributed to the fact that the charge cloud does not instantaneously follow the changes in the nuclear coordinates.

The availability of closely spaced electronic states during the formation of the complex opens up the possibility that sustained low frequency radiation (e.g. in the microwave range) could affect the rate of self-assembly. Sustained radiation would be required since the effect on complex formation would have to build up through the positive feedback relation between the nuclear motions and the spread of electronic states.

Also note that the model requires biological macromolecules to be colorless if their recognition and self-assembly capabilities are to be independent of visible light radiation. If biological materials were colored the superposition of states would be different at night than during the day, for example. To prevent this any chromophores that are present should be segregated from the components of the molecule that make the primary contribution to complex formation. The model suggests that incorporating color groups in a more random manner would alter the rate of complex formation and be incompatible with the discovery of a unique minimum that is independent of ambient light conditions.

9. Conclusions

The seed germination image is intended to capture the context sensitivity of enzymes and other biological macromolecules and to contrast

this to the very limited context sensitivity of electronic switches. The sensitivity is possible because of the ability of enzymes to alter their conformation in a manner that is sensitive to different patterns of milieu influence and to rapidly complex with a specific substrate when appropriate combinations of milieu conditions are present. The seed germination image extends to the self-assembly of polymacromolecular complexes and to the ability of these complexes to adjust their functional activities in a context sensitive way.

The new physical feature of the seed germination model is that the parallelism inherent in the electronic wave function (i.e. the superposition of electronic states) contributes significantly to the context sensitivity, specificity and speed. The parallelism is converted to speed of complex formation through electronic-conformational interactions. The spreading out of electronic states allows the electronic system to explore a potential surface and then drag the nuclear system after it. This is possible because the changing phase relations among different components of the electronic wave function agitate the nuclei, thereby opening up new pathways on the potential surface. The spreading out of the electronic wave function thus serves as a scanning mechanism that is additional to mutual shape exploration through classical Brownian search. Ultimately, of course, this spreading has its origin in heat motion (i.e. derives from the perturbing effect of the infrared radiation field on non-Born-Oppenheimer electrons). The important feature is that it is possible for the electronic spreading and nuclear agitation to self-amplify, leading to the dynamic disappearance of local minima. The picture is clearly quite different than that of molecular dynamics, where quantum mechanics is used only to set up the interactions among the nuclei, which then evolve according to Newton's equation. In our model the electrons lead as well as follow.

The seed germination model implies that the computational and control capabilities of enzymatic networks are very much greater than

would be expected on the basis of mathematical models that are suitable for describing networks of simple elements, such as electronic switches. In principle, it should be possible to simulate enzyme-controlled systems with digital computers, but in practice the computational demands imposed by any adequate model would rapidly outstrip even the most powerful digital computers. The implication is that the 'territory' of biomolecular information processing and control may require a number of complementary mathematical descriptions, each of which maps some abstracted aspect of the territory. By conceptually pasting together these partial maps it should be possible to obtain a useful picture of the whole territory, and therefore of the true capabilities.

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